

REFERENCE MATERIALS

The surface of earth is covered with air layer, which consists of nitrogen(78Vo 1%), oxygen(21Vo 1%) and small amount of CO₂, AR, H₂, Ne, Kt and Xe. To understand air movement and the law of nature, various physical terms as follow should be utilized. For better mutual understanding, scientist in most countries come to same unit system, called SI unit. Following are the relation between generally used engineering terms and new SI unit. In defining aerology, following term and words are required.

BASIC UNITS

Unit	Sign	Eng. unit	SI unit
Length	<i>l</i>	m	m
Mass	m	$\frac{kp.s^2}{m}$	kg
Time	t	Sec.	Sec.
Temp.	T	Degress Celsius(°C)	Kelvin(K)
Electric current	I	Ampere(A)	Ampere(A)
Intensity light	I		Candela(cd)
	n		Mole(mol)

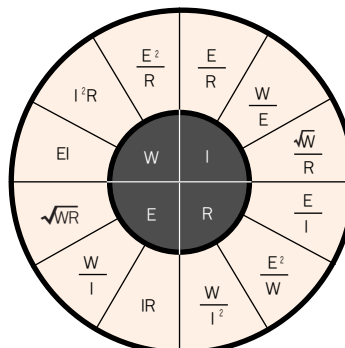
DERIVED UNITS

Unit	Sign	Eng. unit	SI unit
Force	F	kP	Newton(N) $1N = \frac{1kg.m}{s^2}$
Area	A	m ²	m ²
Volume	V	m ³	m ³
Flux	V(Q)	m ² /S	m ² /S
Pressure	P	(Kp/cm ²)	Pascal(Pa) Bar(bar) $1bar = 10^5 Pa = 100kPa(10^2 kPa) = 14.5psi$ $1Pa = \frac{1N}{m^2}$

CONVERSION TABLE OF PRESURE UNITS

No.	Pa	bar	kgf/cm ²	atm	mmH ₂ O	mmHg Torr	psi
1	1	1×10^{-5}	1.01972×10^{-5}	9.86923×10^{-6}	1.01972×10^{-4}	7.50062×10^{-3}	1.45038×10^{-4}
2	1×10^5	1	1.01972	9.86923×10^{-4}	1.01972×10^4	7.50062×10^2	1.45038 × 10
3	9.80665×10^4	9.80665×10^{-4}	1	9.67841×10^{-4}	1×10^4	7.35559×10^2	1.42234 × 10
4	1.01325×10^5	1.01325	1.03323	1	1.03323×10^4	7.60×10^2	1.46960×10
5	9.80665	9.80665×10^{-5}	1×10^{-4}	9.67841×10^{-5}	1	7.35559×10^{-2}	1.42234×10^{-3}
6	1.33322×10^2	1.33322×10^{-3}	1.35951×10^{-3}	1.31579×10^{-3}	1.35951×10	1	1.93368×10^{-2}
7	6.89473×10^3	6.89473×10^{-2}	7.03065×10^{-2}	6.80457×10^{-2}	7.03067×10^2	5.17147×10	1

DIAGRAM FOR THE LAW OF Q



- I = Electric Current (A)
- E = Voltage (V)
- R = Resistance (Ω)
- W = Power consumption (w)

ALLOWABLE TEMPERATURE ACCORDING TO THE TYPE OF INSULATOR

Insulation Type	Temperature rising value (°C)	Max, allowable temperature (°C)
Y Type *	below 50	above 90
A Type	below 65	above 105
E Type	below 80	above 120
B Type	below 90	above 130
F Type	below 115	above 155
H Type	below 140	above 180
C Type *	below 140	above 180

* Theoretical Specifications

CONSUMPTION ELECTRIC POWER

AC Solenoid

electric power(VA) = Voltage(V) × electric current(A)

Consumption electric power(W) =

Voltage(V) × electric current(A) × COS θ

DC Solenoid

Consumption electric power(W) = Voltage(V) × electric current (A)